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Body armor panel for use with personal protective vest and system for assembling same

Abstract

A personal protective vest assembly is described herein. The personal protective vest assembly includes a personal protective vest and a body armor panel positioned within the personal protective vest. The body armor panel includes a ballistic material panel assembly that includes a plurality of layered material segments defined between a strike face and a wear face. Each of the layered material segments includes a different ballistic material.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/282,517 filed on Nov. 23, 2021, the disclosure of which is hereby incorporated by reference in its entirety for all purposes.

FIELD OF THE INVENTION
(1) This invention relates generally to personal protective vests, and more specifically, to a body armor panel for use with a personal protective vest.
BACKGROUND OF THE INVENTION
(2) At least some known personal protective vests include rigid ballistic panels that are sown into the vest to provide the wearer protection from small arms fire. These rigid ballistic panels to not allow for a freedom of movement that is desired by security personal to provide additional comfort and flexibility, while providing a high level of safety protection.
(3) The present invention addresses one or more of the aforementioned challenges.
SUMMARY OF THE INVENTION
(4) In different embodiments of the present invention, a personal protective vest and a body armor

panel for use with the personal protective vest are provided.
(5) In one embodiment of the present invention, a body armor panel for use with a personal protective vest is provided. The body armor panel includes an outer cover defining a panel assembly cavity and a ballistic material panel assembly positioned within the panel assembly cavity. The ballistic material panel assembly includes a plurality of layered material segments defined between a strike face and a wear face. Each of the layered material segments includes a different ballistic material. The plurality of layered material segments includes a first layered material segment including a layer of a first ballistic material defining the strike face, a second layered material segment adjacent the first layered material segment and including a plurality of layers of a second ballistic material, a third layered material segment adjacent the second layered material segment and including a plurality of layers of a third ballistic material, and a fourth layered material segment adjacent the third layered material and defining the wear face.
(6) In another embodiment of the present invention, a personal protective vest assembly is provided. The personal protective vest assembly includes a personal protective vest and a body armor panel positioned within the personal protective vest. The personal protective vest includes an outer surface, an inner surface, and a ballistic panel pocket defined between the outer surface and the inner surface. The inner surface includes a slot for accessing the ballistic panel pocket. The body armor panel is removably positioned within the ballistic panel pocket. The body armor panel includes an outer cover defining a panel assembly cavity and a ballistic material panel assembly positioned within the panel assembly cavity. The ballistic material panel assembly includes a plurality of layered material segments defined between a strike face and a wear face. Each of the layered material segments includes a different ballistic material. The plurality of layered material segments includes a first layered material segment including a layer of a first ballistic material defining the strike face, a second layered material segment adjacent the first layered material segment and including a plurality of layers of a second ballistic material, a third layered material segment adjacent the second layered material segment and including a plurality of layers of a third ballistic material, and a fourth layered material segment adjacent the third layered material segment and defining the wear face.
(7) In a further embodiment, a method of fabricating a body armor panel is provided, The method includes forming a ballistic material panel assembly including a plurality of layered material segments defined between a strike face and a wear face, applying a first heat treatment to a ballistic material panel assembly to increase a temperature of the ballistic material panel assembly to a predefined temperature, and applying a compression treatment to the heated ballistic material panel assembly at a predefined pressure and a predefined period of time to form a hardened ballistic material panel assembly. The method also includes positioning the hardened ballistic material panel assembly within an outer cover and welding the outer cover to form a perimeter seam about a perimeter of the hardened ballistic material panel assembly to seal the hardened ballistic material panel assembly within the outer cover to form a body armor panel. The method also includes applying a second heat treatment to the body armor panel to increase a temperature of the body armor panel to a second predefined temperate and applying a conditioning treatment to the heated body armor panel to increase a flexibility of the body armor panel

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Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Other advantages of the present invention will be readily appreciated as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings wherein:
(2) FIGS. **1-5** are perspective views of a personal protective vest assembly including a personal

protective vest and body armor panels;
(3) FIGS. **6-9** are schematic views of a body armor panel that may be used with the personal protective vest assembly shown in FIG. **1**;
(4) FIGS. **10** and **11** are schematic cross-sectional views of the body armor panel shown in FIGS. **6-9**;
(5) FIG. **12** is a cross-sectional view of the body armor panel shown in FIG. **10**;
(6) FIG. **13** is a partial view of a strike face of the body armor panel shown in FIG. **10**; and
(7) FIG. **14** is a partial view of a wear face of the body armor panel shown in FIG. **10**; and
(8) FIGS. **15** and **16** are flowcharts illustrating a method of manufacturing the body armor panel.
(9) FIG. **17** is a schematic view of a system used to manufacture the body armor panel.
(10) FIGS. **18-21** are schematic views of a brass mold and ballistic material panel assembly used to form the body armor panel.
(11) Corresponding reference characters indicate corresponding parts throughout the drawings.
DETAILED DESCRIPTION OF THE INVENTION
(12) In the following description, numerous specific details are set forth in order to provide a thorough understanding of the present invention. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of various embodiments of the present invention. Also, common but well-understood elements that are useful or necessary in a commercially feasible embodiment are often not depicted in order to facilitate a less obstructed view of these various embodiments of the present invention. It will be apparent, however, to one having ordinary skill in the art that the specific detail need not be employed to practice the present invention. In other instances, well-known materials or methods have not been described in detail in order to avoid obscuring the present invention.
(13) In general, the present invention describes a personal protective vest system that includes a soft-flexible armor panel that is inserted into a personal protective vest. In some embodiments, the personal protective vest system includes the Hyperline™ soft-flexible armor panel provided by Safe Life Defense™.
(14) Referring to FIGS. **1-14**, in the illustrated embodiment, the present invention includes a personal protective vest assembly **10** that includes a wearable personal protective vest **12** and one or more body armor panels **14** inserted into the personal protective vest **12**. In some embodiments, the personal protective vest **12** is configured to be worn by a person covering a portion of the person's torso. In other embodiments, the personal protective vest assembly **10** may include other wearable protective articles that cover other areas of a person's body such as, for example, wearable protective articles covering the lower body, arms, legs, and head.
(15) In the illustrated embodiment, the personal protective vest **12** includes a front vest panel **16** and a rear vest panel **18**. In some embodiments, the front vest panel **16** may be removable coupled to the rear vest panel **18** with a vest fastening assembly **20**. The vest fastening assembly **20** may include, for example, a Velcro assembly, a button assembly, and/or any suitable fasteners that allow the front vest panel **16** to removable attach to the rear vest panel **18**.
(16) Each vest panel **16**, **18** includes an outer surface **22** and an inner surface **24** adapted to be positioned adjacent to the person's body with the personal protective vest **12** worn by the person. A ballistic panel pocket **26** is defined between the outer surface **22** and the inner surface **24**. The ballistic panel pocket **26** is sized and shaped to receive a body armor panel **14** therein such that the body armor panel **14** is positioned within the ballistic panel pocket **26**. The inner surface **24** also including a slot **28** defined along the inner surface **24** to provide access to the ballistic panel pocket **26** to insert and/or remove body armor panel **14**. A slot fastening assembly **30** may be defined along the slot **28** to allow the ballistic panel pocket **26** to be selectively sealed. The slot fastening assembly **30** may include, for example, a Velcro assembly, a button fastener assembly, and/or any suitable fasteners that allow the slot **28** and ballistic panel pocket **26** to be sealed.
(17) In the illustrated embodiment, the body armor panel **14** includes an outer cover **32** that defines

a panel assembly cavity **34** and a ballistic material panel assembly **36** that is positioned within the panel assembly cavity **34**. The outer cover **32** may include a 30 denier thermoplastic polyurethane laminated nylon fabric. In some embodiments, the body armor panel **14** includes a cross-sectional shape substantially similar to the cross-sectional shape of the corresponding vest panels **16**, **18**. In other embodiments, the body armor panel **14** may include any suitable cross-sectional shape to facilitate covering any portion of a person's body.

(18) The ballistic material panel assembly **36** includes a plurality of layered material segments **38**, **40**, **42**, **44** defined between a strike face **46** and a wear face **48**, with each of the layered material segments **38**, **40**, **42**, **44** including a different ballistic material. When positioned within the ballistic panel pocket **26** the strike face **46** is orientated near the outer surface **22** of the corresponding vest panel **16**, **18** and the wear face **48** is orientated near the inner surface **24** of the corresponding vest panel **16**, **18** such that the wear face **48** is closer to a person's body than the strike face **46** with the personal protective vest **12** worn by the person.

(19) In the illustrated embodiment, the plurality of layered material segments includes a first layered material segment **38** that includes a layer of a first ballistic material **50**, a second layered material segment **40** that is adjacent the first layered material segment **38** and includes a plurality of layers of a second ballistic material **52**, a third layered material segment **42** that is adjacent the second layered material segment **40** and includes a plurality of layers of a third ballistic material **54**, and a fourth layered material segment **44** that is adjacent the third layered material segment **42** and includes a plurality of layers of a fourth ballistic material **56**. The first layered material segment **38** defines the strike face **46** and the fourth layered material segment **44** defines the wear face **48**. A number of layers of the third ballistic material **54** is greater than a number of layers of the second ballistic material **52**, and a number of layers of the fourth ballistic material **56** is less than the number of layers of the third ballistic material **54**.

(20) In some embodiments, as shown in FIG. **10**, the first layered material segment **38** includes a single layer of the first ballistic material **50**, the second layered material segment **40** includes four layers of the second ballistic material **52**, the third layered material segment **42** includes eight layers of the third ballistic material **54**, and the fourth layered material segment **44** includes four layers of the fourth ballistic material **56**. In other embodiments, each segment **38**, **40**, **42**, **44** may include any suitable number of layers of corresponding ballistic material.

(21) The first ballistic material **50** includes an ultra-high molecular weight polyethylene fiber based composite laminate having an Areal density of about between 226 g/m.sup.2 and 240 g/m.sup.2 including four single layers of unidirectional sheet cross plied at 90 degrees to each other and consolidated with a rubber based matrix. In some embodiments, the first ballistic material may include DSM Dyneema® HB50 with the first layered material segment **38** includes a single layer of DSM Dyneema® HB50.

(22) The second ballistic material **52** includes an ultra-high molecular weight polyethylene fiber based composite laminate having an Areal density of about between 208 g/m.sup.2 and 224 g/m.sup.2 including six single layers of unidirectional sheet cross plied at 90 degrees to each other and consolidated with a rubber based matrix. In some embodiments, the second ballistic material may include DSM Dyneema® SB117 with the second layered material segment **40** including four layers of DSM Dyneema® SB117.

(23) The third ballistic material **54** includes para-aramid unidirectional material including four plies of para-aramid unidirectional material cross-plied in 0°/90°/0°/90° configuration. In some embodiments, the third ballistic material **54** may include Barrday Advanced Material Solutions™ U611 with the third layered material segment **42** including eight layers of Barrday Advanced Material Solutions™ U611.

(24) The fourth ballistic material **56** includes a woven and laminated material having an Areal density of about 516 g/m.sup.2. In some embodiments, the fourth ballistic material **56** may include Barrday Advanced Material Solutions™ FCKER1017160-01129 with the fourth layered material

segment **44** including four layers of Barrday Advanced Material Solutions™ FCKER1017160-01129.

(25) As shown in FIG. **11**, in some embodiments, the plurality of layered material segments includes the first layered material segment **38** including a single layer of the first ballistic material **50**, the second layered material segment **40** including six layers of the second ballistic material **52**, the third layered material segment **42** including six layers of the third ballistic material **54**, and the fourth layered material segment **44** including a plurality of layers of the third ballistic material **54** and a plurality of layers of the fourth ballistic material **56** positioned in an alternating layered arrangement such that each layer of the third ballistic material **54** is positioned between corresponding layers of the fourth ballistic material **56**. For example, the fourth layered material segment **44** may include four layers of the fourth ballistic material **56** interlaced with three layers of the third ballistic material **54** such that a top layer of fourth ballistic material **56** is adjacent a bottom layer of third ballistic material **54** of the third layered material segment **42**, and a bottom layer of fourth ballistic material **56** defines the wear face **48**.

(26) In some embodiments, as shown in FIGS. **6-8**, the body armor panel **14** includes a cross-sectional shape defining a lower portion **70** and an upper portion **72** extending from the lower portion **70**. The upper portion **72** has a first width **74**, and the lower portion **70** has a second width **76** that is larger than the first width **74** of the upper portion **72**. The body armor panel **14** also has a pair of opposing side walls **78**, **80** that extend from the lower portion **70** to the upper portion **72** at opposing oblique angles. The upper portion **72** also includes an arcuate surface **82** defined along the a top edge **84** of the upper portion **72**. In other embodiments, as shown in FIG. **9**, the body armor panel **14** may include a substantially rectangular cross-sectional shape.

(27) FIG. **15** is a flowchart illustrating a method **200** of manufacturing the body armor panel **14**. The method includes a plurality of steps. Each method step may be performed independently of, or in combination with, other method steps. In the illustrated embodiment, the method **200** includes positioning the ballistic material panel assembly **36** within a dual platen heated press including placing the layered material segments **38**, **40**, **42**, **44** within the dual platen heated press such that the first layered material segment **38** defines the strike face **46**, the second layered material segment **40** is between the first layered material segment **38** and the third layered material segment **42**, and the fourth layered material segment **44** is adjacent the third layered material segment **42** and defines the wear face **48**.

(28) The dual platen heated press is then operated to apply a heat process to the ballistic material panel assembly **36** including applying heat at a first temperature from the strike face **46**, applying heat at a different second temperature from the wear face **48**, and applying pressure to the ballistic material panel assembly **36** during the heat process. In some embodiments, the first temperature is between about 250 and 300° Fahrenheit and the second temperature is between about 150 and 225° Fahrenheit, with the applied pressure at about 80 psi.

(29) The ballistic material panel assembly **36** is then removed from the dual platen heated press after a first predefined elapsed period of time. For example, in some embodiments, the first predefined elapsed period of time is about 1.5 minutes. The ballistic material panel assembly **36** is allowed to cool for a second elapsed period of time and then cut to a desired shaped. In some embodiments, the second predefined elapsed period of time is about 30 seconds.

(30) In some embodiments, the Hyperline™ soft armor panel is constructed by layering ballistic material in a 0°/0° orientation in a layered material configuration from strike face to wear face including: 1 Layer—DSM Dyneema® HB50; 4 Layers—DSM Dyneema® SB117; 8 Layers—Barrday Advanced Material Solutions™ U611; and 4 Layers—Barrday Advanced Material Solutions™ FCKER1017160-01129. The layered materials are feed into a dual platen heated press that is used to apply heat at a temperature of 250-300 degrees Fahrenheit from the strike face layer. A temperature of 150-225 degrees Fahrenheit is applied from the opposite platen to the wear face layer. Pressure is applied during the heat process at no less than 80 psi. After a minimum of 1.5

minutes the press is released and the materials is removed from the machinery and is allowed to cure for a period of no less than 30 seconds. The armor packet may be consolidated in a full sheet from which shapes may be cut by using a multi ply cutter, precision rotary blade or die cutter, or, layers may be pre-cut into uniform shapes that are layered and consolidated using the above process.

(31) The ballistic material panel assembly **36** is sealed inside an outer cover **32** including water resistant TPU material such as, for example, Brookwood™ Cover HST 30d. The TPU is sealed using a High Frequency Ultrasonic Welding Machine. In some embodiments, the body armor panel **14** includes a High Frequency Weld (HFW) seal **58** about the perimeter of the ballistic material panel assembly **36** having a width of about 0.25 inches+/−0.25 inches, with a maximum gap **60** between the ballistic panel **36** and the HFW seal **58** of about 0.25 inches+−0.2 inches, with an overcut **62** having a maximum allowable width equal to 0.5 inches. A thermal label with an adhesive backing is then applied to the wear face of the sealed ballistic panel.

(32) FIG. **16** is a flowchart illustrating another method **300** of manufacturing the body armor panel **14**. FIG. **17** is a schematic view of a system **100** used to manufacture the body armor panel **14**. In the illustrated embodiment, the system **100** includes a fabric spreading and cutting machine **102**, a first industrial conveyor oven machine **104**, a hydraulic press machine **106**, a high frequency ultrasonic welding machine **108**, a second industrial conveyor oven machine **110**, and a belt press machine **112**. In method step **302**, the plurality of layered material segments are fed through the fabric spreading and cutting machine **102** and cut to predefined cross-sectional shapes to form a pre-cut, layered ballistic material panel assemblies **36**.

(33) In method step **304**, a first heat treatment is applied to a pre-cut, layered ballistic material panel assembly **36** to increase a temperature of the ballistic material panel assembly to a predefined temperature. The first heat treatment may be applied to increase the temperature of the ballistic material panel assembly to the predefined temperature between about 150° and 190° Fahrenheit. For example, the pre-cut, layered ballistic material panel assembly **36** may be positioned within an industrial heat tunnel, i.e., the first industrial conveyor oven machine **104**, for a predefined period of time to increase the temperature of the ballistic material panel assembly to the predefined temperature between about 150° and 190° Fahrenheit. In some embodiments, applying the first heat treatment includes operating the industrial heat tunnel at an internal temperature between about 300° and 350° Fahrenheit, and positioning the ballistic material panel assembly within the heat tunnel for an elapsed heating time between about 1 minute and 2 minutes. For example, the first heat treatment may include feeding the pre-cut, layered ballistic material panel assembly **36** through the first industrial conveyor oven machine **104** operated with a heated environment of about 335° F. for 1 minute and 50 seconds. In some embodiments, a reflective adhesive vinyl sticker **86** is applied onto a top layer (i.e., the strike face **46**) of the ballistic material panel assembly **36** prior to the first heat treatment.

(34) In method step **306**, a compression treatment is applied to the heated ballistic material panel assembly at a predefined pressure and a predefined period of time to form a hardened ballistic material panel assembly. For example, upon completion of the first heat treatment, the heated ballistic material panel assembly is removed from the first industrial conveyor oven machine **104** and positioned onto the hydraulic press machine **106**. The hydraulic press machine **106** is then operated to apply the compression treatment including applying a pressure between about 100 to 500 tons to the heated ballistic material panel assembly. The compression treatment may include, for example, applying a pressure between about 100 to 500 tons to the heated ballistic material panel assembly for a period of 30 seconds. In some embodiments, the compression treatment may include applying 200 tons of pressure to the heated ballistic material panel assembly for a period of 30 seconds to form a hardened ballistic material panel assembly.

(35) In embodiments in which the reflective adhesive vinyl sticker **86** has been applied onto the strike face **46** of the ballistic material panel assembly **36** prior to the first heat treatment, the

compression treatment embeds the reflective adhesive vinyl sticker **86** into the strike face **46** such that surface of reflective adhesive vinyl sticker **86** is substantially flush with the strike face **46** of the hardened ballistic material panel assembly.

(36) In method step **308**, the hardened ballistic material panel assembly **36** is positioned within the outer cover **32** and the outer cover **32** is welded to form a perimeter seam about a perimeter of the hardened ballistic material panel assembly to seal the hardened ballistic material panel assembly within the outer cover to form a body armor panel **14**. For example, as shown in FIGS. **16-21**, the upon completing the compression treatment, the hardened ballistic material panel assembly **36** may be positioned between two layers of pre-cut outer cover material (e.g., Brookwood™ Cover HST 30d) having a cross-sectional shape substantially similar to the hardened ballistic material panel assembly **36**, and placed on a bottom plate **114** of the high frequency ultrasonic welding machine **108**. A brass mold assembly **116** having cross-sectional shape substantially similar to the hardened ballistic material panel assembly **36** is positioned over a top layer of the pre-cut outer cover material such that the hardened ballistic material panel assembly **36** is positioned within a cavity **118** defined through the brass mold assembly **116**. A compressible insert **120** is then placed within the cavity **118** on top of the hardened ballistic material panel assembly **36** and outer cover **32**. The compressible insert **120** includes a thickness **122** such that a top surface **124** of the compressible insert **120** is spaced a vertical distance **126** above an upper surface **128** of the brass mold assembly **116** with the compressible insert **120** contacting the outer cover **32**. As shown in FIG. **21**, as a top plate **130** of the high frequency ultrasonic welding machine **108** is moved to contact the brass mold assembly **116**, the top plate **130** contacts the compressible insert **120** to cause the compressible insert **120** to compress and apply pressure onto the hardened ballistic material panel assembly **36** and outer cover **32** to force air trapped between the hardened ballistic material panel assembly **36** and outer cover **32** towards the perimeter of the outer cover **32** before the top plate **130** contacts the brass mold assembly **116**. Once the top plate **130** is in contact with the brass mold assembly **116**, the high frequency ultrasonic welding machine **108** is operated to weld a perimeter seam along the outer cover **32** via the brass mold assembly **116** to form a vacuum seal between the hardened ballistic material panel assembly **36** and the outer cover **32**.

(37) In some embodiments, the high frequency ultrasonic welding machine **108** is operated to seal the hardened ballistic material panel assembly **36** within a water resistance thermo-plastic packaging. The water resistance thermo-plastic packaging may include a semi-transparent thermo-plastic material such that the reflective adhesive vinyl sticker **86** is visible through the water resistance thermo-plastic outer cover. In some embodiments, the compressible insert **120** includes a Nomex® material positioned within the brass mold assembly **116**. As the high frequency ultrasonic welding machine **108** operates the top plate **130** to lower and contact the brass mold assembly **116**, the top plate **130** contacts the Nomex insert to apply pressure to the top layer of water-resistance material to remove air between top/bottom layers of material and the hardened ballistic material panel assembly **36** to facilitate eliminating air from the body armor panel **14**. The high frequency ultrasonic welding machine **108** then applies high frequency radio waves to water resistance material layers through brass mold assembly to form the perimeter weld to seal the hardened ballistic material panel assembly **36** within the water resistant outer cover packaging and form an air-tight seal. The air-tight seal facilitates maintaining the shape of the assembled panel.

(38) In another embodiment, the first heat treatment and the compression treatment are applied to only a portion of the plurality of the plurality of layered material segments **38**, **40**, **42**, **44**. For example, in some embodiments, the first heat treatment and the compression treatment are applied only to the first and second layered material segments **38**, **40** to form a partially hardened ballistic material panel assembly. The partially hardened ballistic material panel assembly is then layered onto the third layered material segment **42** and/or the fourth layered material segment **44** to form the completed ballistic material panel assembly, which is then placed within the outer cover **32** and into the high frequency ultrasonic welding machine **108** to weld the perimeter seam about the

completed ballistic material panel as sembly.
(39) In other embodiments, the first heat treatment and the compression treatment are applied only to the third layered material segment **42** and/or the fourth layered material segment **44** to form the partially hardened ballistic material panel assembly. The first and second layered material segments **38**, **40** are then layered on top of the partially hardened ballistic material panel assembly to form the completed ballistic material panel assembly prior to welding the outer cover **32** onto the completed ballistic material panel assembly.
(40) In method step **310**, a second heat treatment is applied to the body armor panel **14** to increase a temperature of the body armor panel **14** to a second predefined temperate. For example, after the hardened ballistic material panel assembly **36** has been welded within the outer cover **32** to form the sealed body armor panel **14**, the sealed body armor panel **14** is then placed into the second industrial conveyor oven machine **110** for a predefined period of time to increase the temperature of the sealed body armor panel **14** to facilitate increasing a flexibility of the sealed body armor panel **14**. The second industrial conveyor oven machine **110** is operated at a lower internal temperature then the first industrial conveyor oven machine **104**.
(41) In method step **312**, a conditioning treatment is then applied to the heated body armor panel **14** to increase a flexibility of a conditioned body armor panel **14**. For example, after the second heat treatment is completed, the heated body armor panel **14** is fed through a belt press machine **112** that includes a pair of opposing conveyor belt systems that form roller pairs of varying heights. As the heated body armor panel **14** is moved through the roller pairs the heated body armor panel **14** is moved through various vertical positions, thereby breaking-in the heated body armor panel **14** to increase the flexibility of the finished, conditioned body armor panel **14**.
(42) Obviously, many modifications and variations of the present invention are possible in light of the above teachings. The invention may be practiced otherwise than as specifically described within the scope of the appended claims.
(43) Although specific features of various embodiments of the disclosure may be shown in some drawings and not in others, this is for convenience only. In accordance with the principles of the disclosure, any feature of a drawing or other embodiment may be referenced and/or claimed in combination with any feature of any other drawing or embodiment.
(44) This written description uses examples to describe embodiments of the disclosure and also to enable any person skilled in the art to practice the embodiments, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

Claims

1. A body armor panel for use with a personal protective vest, comprising: an outer cover defining a panel assembly cavity; and a flexible ballistic material panel assembly positioned within the panel assembly cavity, the flexible ballistic material panel assembly including a plurality of layered material segments defined between a strike face and a wear face, each of the layered material segments including a different ballistic material, the plurality of layered material segments including: a first layered material segment including a layer of a first ballistic material defining the strike face, the first ballistic material is formed of an ultra-high molecular weight polyethylene fiber based composite laminate having an Areal density of about between 226 g/m2 and 240 g/m2 including four single layers of unidirectional sheet cross plied at 90 degrees to each other and consolidated with a rubber based matrix; a second layered material segment adjacent the first layered material segment and including a plurality of layers of a second ballistic material that is

different than the first ballistic material, the second ballistic material is formed of an ultra-high molecular weight polyethylene fiber based composite laminate having an Areal density of about between 208 g/m2 and 224 g/m2 including six single layers of unidirectional sheet cross plied at 90 degrees to each other and consolidated with a rubber based matrix; and a third layered material segment adjacent the second layered material segment and including a plurality of layers of a third ballistic material that is different than the first and second ballistic materials; and a fourth layered material segment adjacent the third layered material segment and defining the wear face, the fourth layered material segment including a plurality of layers of a fourth ballistic material that is different than the third ballistic material.

2. The body armor panel of claim 1, wherein the fourth layered material segment includes a plurality of second layers of the third ballistic material and a plurality of layers of a fourth ballistic material that is different than the third ballistic material, wherein each second layer of the third ballistic material is positioned between corresponding layers of the fourth ballistic material in an alternating layered arrangement.

3. The body armor panel of claim 1, wherein a number of layers of the fourth ballistic material is less than the number of layers of the third ballistic material.

4. The body armor panel of claim 1, wherein the first layered material segment includes a single layer of the first ballistic material, the second layered material segment includes four layers of the second ballistic material, the third layered material segment includes eight layers of the third ballistic material, and the fourth layered material segment includes four layers of the fourth ballistic material.

5. The body armor panel of claim 1, wherein the first ballistic material includes Dyneema® HB50.

6. The body armor panel of claim 5, wherein the second ballistic material includes Dyneema® SB117.

7. The body armor panel of claim 1, wherein the third ballistic material includes para-aramid unidirectional material.

8. The body armor panel of claim 7, wherein the third ballistic material includes four plies of para-aramid unidirectional material cross-plied in 0°/90°/0°/90° configuration.

9. The body armor panel of claim 1, wherein the fourth ballistic material includes a woven and laminated material having an Areal density of about 516 g/m2.

10. The body armor panel of claim 1, wherein the outer covering includes a 30 denier thermoplastic polyurethane laminated nylon fabric.

11. A personal protective vest assembly comprising: a personal protective vest including an outer surface, an inner surface, and a ballistic panel pocket defined between the outer surface and the inner surface, the inner surface including a slot for accessing the ballistic panel pocket; and body armor panel removably positioned within the ballistic panel pocket, the body armor panel including: an outer cover defining a panel assembly cavity; and a flexible ballistic material panel assembly positioned within the panel assembly cavity, the flexible ballistic material panel assembly including a plurality of layered material segments defined between a strike face and a wear face, each of the layered material segments including a different ballistic material, the plurality of layered material segments including: a first layered material segment including a layer of a first ballistic material defining the strike face, the first ballistic material is formed of an ultra-high molecular weight polyethylene fiber based composite laminate having an Areal density of about between 226 g/m2 and 240 g/m2 including four single layers of unidirectional sheet cross plied at 90 degrees to each other and consolidated with a rubber based matrix; a second layered material segment adjacent the first layered material segment and including a plurality of layers of a second ballistic material, the second ballistic material is formed of an ultra-high molecular weight polyethylene fiber based composite laminate having an Areal density of about between 208 g/m2 and 224 g/m2 including six single layers of unidirectional sheet cross plied at 90 degrees to each other and consolidated with a rubber based matrix; and a third layered material segment adjacent

the second layered material segment and including a plurality of layers of a third ballistic material that is different than the first and second ballistic materials; and a fourth layered material segment adjacent the third layered material segment and defining the wear face, the fourth layered material segment including a plurality of layers of a fourth ballistic material that is different than the third ballistic material.